Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

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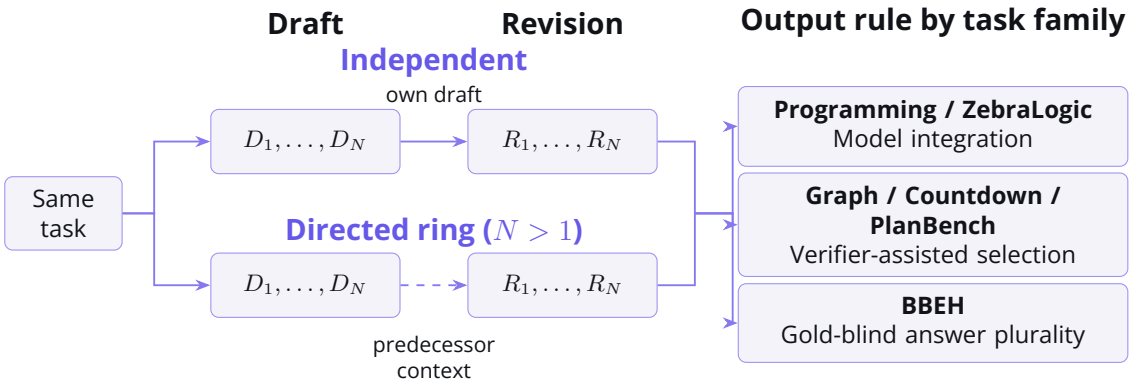
Abstract

Adding workers changes token use, worker participation and answer selection as well as coordination. We examine eleven scaling studies with 20,376 evaluation trials across seven task families and 6,272 calibration trials, using Luna and Muse Spark at requested widths up to sixteen. All 148 wider configurations use more recorded tokens than their matched-ceiling single-worker controls: 75 score lower, 24 tie and 49 improve. Budget-dependent feasibility is visible on PlanBench: independent four-worker accuracy rises from 0% to 100% when the ceiling doubles, while truncated stages fall from 99.94% to 0.19%. On BigCodeBench, Muse's sixteen-worker activation ranges from 7.3% to 93.5% across three ceilings. A fresh 300-task ZebraLogic confirmation transfers a communicating 4/8/16 policy without refitting, but its advantage over fixed four remains uncertain (+0.92 percentage points; 95% CI $[-0.58, 2.50]$). Ring revision provides no consistent advantage over independent revision. These heterogeneous studies use task-paired analyses, study-specific output rules and amended operational protocols. Recorded token debit does not measure exact total compute or retry expenditure. The findings motivate evaluating allocation policies with execution records and competitive controls; equal realized compute advantages and general worker-scaling laws remain unestablished.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

"All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).

Adapted to this study: All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.



Nominal stage shares: programming / ZebraLogic: 50% draft, 25% revision, 25% integration; later graph / Countdown / BBEH / PlanBench diagnostics: 50% draft, 50% revision.

Figure 1 | Study-specific execution. Independent revision uses its own draft; ring revision receives predecessor context. Both conditions share the study's output rule. Requested width N can exceed active participation when prompt reserves prevent calls. Tested widths are 1/2/4/8/16 where applicable; nominal ceilings and retained token debit are distinct.

1 Three Checks for Scaling Claims

Width scaling evaluates a resource-allocation policy as well as coordination (Snell et al., 2024; Kim et al., 2026). From eleven studies, we derive three reporting checks: stage completion, realized worker participation and a competitive fixed-width control.

Design. We use BigCodeBench, MBPP/EvalPlus, ZebraLogic, BBEH and PlanBench subsets (Zhuo et al., 2025; Austin et al., 2021; Liu et al., 2023; Lin et al., 2025; Kazemi et al., 2025; Valmeekam et al., 2023), and custom graph/Countdown tasks with deterministic checks; graph generation is inspired by Reasoning Gym (Stojanovski et al., 2025). Widths reach sixteen in coding and ZebraLogic; diagnostics test one and four. Models and routing vary; Figure 1 shows output rules. Verifier-assisted gains include selection.

Stage completion. Doubling PlanBench's ceiling reverses the independent $N = 4$ effect, from -95.0 pp (CI $[-98.0, -91.5]$) to $+10.5$ pp (CI $[6.0, 15.5]$). Higher-ceiling debit is 4.06 times the single-worker control; stage truncation falls from 99.94% to 0.19%. Figure 2 summarizes 148 wider cells sharing tasks and controls.

Active participation. BigCodeBench Muse's $N = 16$ activation is 7.3%, 45.3% and 93.5% at ceilings 32,768, 65,536 and 131,072. At 32,768, independent widths 1/8/16 pass 40/64, 36/64 and 14/64. Activation's causal role remains unresolved.

Competitive baseline. On 300 fresh ZebraLogic puzzles, transferred ring $4/8/16$ minus fixed four averages $+0.92$ pp at 65,536/131,072 tokens ($[-0.58, 2.50]$), establishing neither superiority nor equivalence. At 131,072, sixteen communicating workers score 90.33% versus four's 89.67%, using 3.57 times retained debit.

Attributing communication gains. Sampling, aggregation and debate are distinct (Brown et al., 2024; Wang et al., 2023; Du et al., 2023). Countdown's independent gains of 9.5–17.0 pp use 2.77–3.85 times recorded tokens; its pooled ring effect is -5.0 pp ($[-8.5, -1.75]$). Seven Muse graph zeros are operational failures, not necessarily reasoning errors. These ring results have narrower scope than shared-workspace communication (Park et al., 2026); execution audits complement failure taxonomies (Cemri et al., 2025).

Uncertainty and limits. Task bootstrap retains configurations and repetitions, stratified where applicable (Efron, 1979; Field and Welsh, 2007); 95% CIs lack cross-study multiplicity correction. Pilots and smoke checks are excluded. Debit omits retry expenditure and can conservatively charge unknown usage; equal realized compute is unestablished.

Practical implication. Report stage completion and active workers alongside accuracy, and compare transferred policies with a competitive fixed width. Whether coordination helps at matched realized compute remains open.
Study records: https://github.com/PluvioXO/agent-swarms-reproducibility.

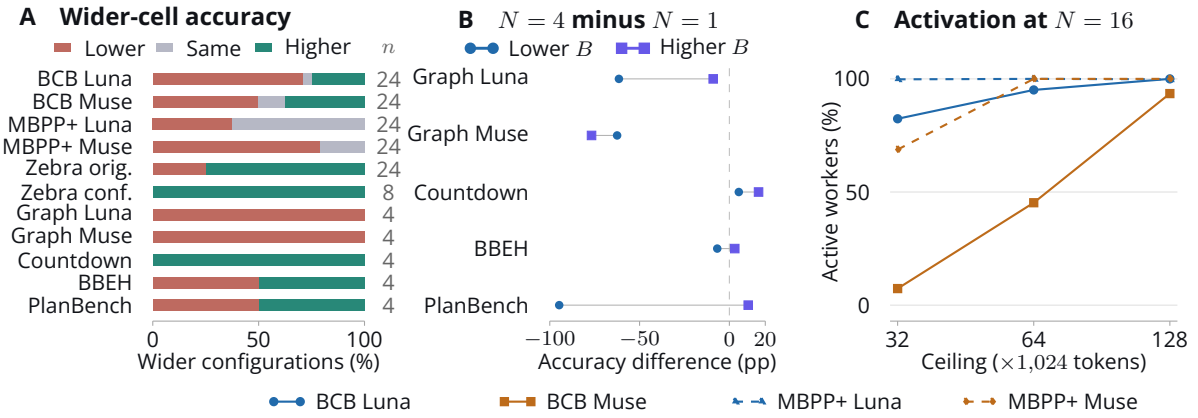


Figure 2 | Evaluation results. (A) Wider cells versus matched-ceiling $N = 1$; n counts configurations. (B) Four-worker effects pooled over revision conditions, at ceilings 4/8k for graphs and 8/16k otherwise ($k = 1{,}024$ tokens). (C) Activation at $N = 16$.

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